



Effects of replacing canola meal with extruded soybean meals on lactational performance and enteric gas emissions in dairy cows

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ABSTRACT

Mechanically extracting oil through an extrusion process increases rumen-undegraded protein content of oilseed meals. This study investigated the effects of replacement on an equal CP basis of canola meal (CM) with extruded soybean meals (ESBM) processed at 2 extruder temperatures, 149°C (LSBM) and 171°C (HSBM), on lactational performance and enteric gas emissions of dairy cows. Following a 2-wk covariate period, 45 Holstein cows averaging ($\pm$ SD): 117 $\pm$ 64 DIM and 45 $\pm$ 8 kg/d milk yield (MY) were assigned to 1 of 3 treatments in a 9-wk randomized complete block design experiment. Cows were allowed the first 3 wk for adaptation to treatments and the last 6 wk were for data and sample collection. Treatments contained: 13.7% CM (with 42.6% CP; CM diet), 12.1% LSBM (with 47.8% CP; LSBM diet), and 12.5% HSBM (with 46.1% CP; HSBM diet). Enteric CH₄ emission was measured using the GreenFeed system. Orthogonal contrasts were used to evaluate the effect of replacing CM with ESBM and source of ESBM (i.e., LSBM vs. HSBM). Estimated MP supply was 9% below requirements for cows fed CM whereas MP requirements of the cows were met by the ESBM diets. Dry matter intake, ECM, and ECM feed efficiency (ECM/DMI) did not differ among treatments. Milk urea N was lower and fat concentration was higher in cows fed CM compared with ESBM. When compared with ESBM, milk fat yield and milk true protein (MTP) content tended to be increased in cows fed the CM diet. In addition, MTP tended to be decreased in cows fed HSBM, compared with LSBM. Daily CH₄ emission and emission intensity and yield were not affected by diet. Ruminal molar proportions of propionate decreased, whereas molar proportions of acetate and acetate-to-propionate ratio increased, and isobutyrate and valerate tended to decrease with HSBM, compared with the LSBM diet. Total-tract digestibility of

nutrients was not different between the CM and ESBM diets. When compared with cows fed the LSBM diet, apparent total-tract digestibility of DM, OM, CP, and starch were decreased in cows fed the HSBM diet. Urinary urea N excretion was greater in cows fed ESBM relative to CM, whereas fecal N excretion was increased by the HSBM diet compared with LSBM. This study indicates that substituting CM with ESBM on an equal CP basis does not alter lactational performance in dairy cows. Additionally, using ESBM produced at higher extrusion temperatures offered no apparent advantage in terms of lactational performance of dairy cows.

Key words: extruded soybean meal, extrusion temperature, canola meal, milk production

INTRODUCTION

Processing of canola seeds for oil production has been increasing in recent years, thus promoting the production of its byproduct, canola meal (CM; Canola Council of Canada, 2024). Consequently, CM has become a widely used source of protein in dairy cattle diets. In contrast, soybean meal (SBM), a product of the soybean oil industry, has historically been the primary protein concentrate in dairy cattle diets (Huhtanen et al., 2011). Research has shown that substituting solvent-extracted soybean meal (SSBM) with CM increases DMI and milk yield (MY; Huhtanen et al., 2011; Martineau et al., 2013). However, CM degradable protein fractions are more similar to that of extruded soybean meal (ESBM) than to SSBM (Lage et al., 2021; Cueva et al., 2023b). The discrepancy originates from the processing methods used during production of CM and SBM, which significantly influence their protein fractions. For instance, during the production of CM, the high oil content of canola seeds requires a mechanical extrusion process before solvent extraction (Canola Council of Canada, 2024). Extrusion involves forcing the seeds through a die to extract the oil, which creates friction and high temperatures (Björck et al., 1984), thus increasing the RUP content of the meal (Giallongo et al., 2015). Therefore, comparing CM di-

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rectly to SSBM seems to be misleading because it does not account for differences in protein fractions and RUP between the feeds, and the extrusion process results in CM having similar RUP content to that of ESBM (Maxin et al., 2013a; Lage et al., 2021; Cueva et al., 2023b).

Despite similarities in RUP content between CM and ESBM, ESBM has superior intestinal digestibility of RUP compared with CM (Lage et al., 2021; NASEM, 2021; Cueva et al., 2023b). Therefore, in studies where CM was replaced with ESBM on an equal CP basis, the ESBM diets provided a greater MP supply for the cows (Lage et al., 2021; Cueva et al., 2023b); however, the additional MP provided by ESBM, as compared with CM, was not used for production purposes. This led us to hypothesize that the high overall dietary MP supply in these studies may have masked the expected effects on milk production originating from differences in RUP and EAA supplies between CM and ESBM. Consequently, it is plausible that high-producing cows may benefit from increased escape protein supply from ESBM relative to CM when MP supply is below requirements. In addition, CM and ESBM also largely differ in their oil and NDF contents. Previous analyses have indicated that increasing dietary ether extract (EE) can result in reduced CH₄ production (Eugène et al., 2008; Martins et al., 2024). Therefore, the inherently higher fat content in ESBM compared with CM could potentially decrease enteric CH₄ emissions both in absolute terms and relative to feed intake and milk production of the cows.

Amino acids for dairy cows primarily originate from microbial protein synthesized in the rumen and digested feed RUP (NASEM, 2021). The RUP content of ESBM increases linearly with extrusion temperature (Isenberg et al., 2012; Giallongo et al., 2015). Consequently, increasing the extrusion temperature during the production of ESBM, leading to higher RUP content in the meal, may enhance its relative feeding value compared with CM or ESBM produced at lower extrusion temperatures. Therefore, it is plausible that cows could experience further benefits from the increased supply of digestible RUP from ESBM processed at higher temperatures.

To the best of our knowledge, there are no long-term experiments assessing the effects of substituting CM with ESBM processed at different extrusion temperatures. Therefore, this study was designed to investigate the effects of equal CP replacement of CM with ESBM produced using 2 different temperatures on lactational performance, rumen fermentation, enteric gas emissions, nutrient digestibility, N excretions, and milk fatty acid (FA) profile in mid-lactation dairy cows. A secondary objective of the experiment was to determine whether increasing the extrusion temperature during ESBM production could elicit lactation performance benefits by increasing rumen escape protein supply. Our hypothesis

was that substituting CM with ESBM, with greater feed RUP, on an equal CP basis in MP-deficient diets would be of benefit to the cow, thereby improving the lactational performance of dairy cows. Additionally, due to its higher oil content, we hypothesized that ESBM may reduce enteric CH₄ emissions compared with CM.

MATERIALS AND METHODS

All procedures involving animals carried out in the experiment were reviewed and approved by The Pennsylvania State University's Animal Care and Use Committee.

Animals and Diets

The study was conducted with a total of 45 primi- (12) and multiparous (33) Holstein cows averaging ($\pm$ SD): 117 ± 64 DIM and 45 ± 8 kg/d MY at the beginning of the experiment. Nine ruminally cannulated cows were selected at the beginning of the study for collection of rumen samples; however, a in the HSBM treatment group died before the ruminal sample collection period. Therefore, a total of 8 multiparous rumen-cannulated cows were used in the experiment for collection of rumen samples (further details are included in the "Ruminal Fermentation" section). Cows were housed at The Pennsylvania State University's Dairy Teaching and Research Center freestall barn, equipped with a Calan Broadbent Feeding System (American Calan Inc., Northwood, NH) for individual monitoring of cow DMI.

The experiment lasted a total of 11 wk and was a randomized complete block design experiment with a 2-wk covariate period at the beginning of the study, followed by a 9-wk experimental period, with the first 3 wk used for adaptation to treatments and the last 6 wk for data and sample collection. We refer to the week during which data and samples were collected as experimental wk 1 to 6. Cows were blocked into 15 blocks of 3 cows each based on parity, DIM, and MY during the covariate period. Cows had free access to drinking water, and diets were fed as TMR using a Rissler model 1050 TMR mixer (I. H. Rissler Mfg. LLC). Feeding was once daily at ~0800 h, and feed was offered for ad libitum intake to ~10% refusals. The cows were milked twice daily at ~0645 and 1745 h.

Cows within block were assigned randomly to 1 of 3 treatments. Treatment diets contained: 13.7% CM (CM diet; 42.6% CP), 12.1% ESBM produced at 149°C (i.e., low-temperature, **LSBM** diet; 47.8% CP), and 12.5% ESBM produced at 171°C (i.e., high-temperature, **HSBM** diet; 46.1% CP). Because ESBM has higher a CP content than CM, soybean hulls were added to the ESBM diets to offset the difference in inclusion rate between CM and ESBM and partially compensate for the greater

Table 1. Ingredient and nutrient composition of the diets used in the experiment

Item	Diet ¹		
	CM	LSBM	HSBM
Ingredient, % of DM or as indicated			
Corn silage ²	38.2	38.3	38.0
Alfalfa haylage ³	20.1	20.2	20.0
Straw-hay mixture ⁴	0.50	0.50	0.50
Cottonseed, whole	8.53	8.55	8.50
Corn grain, ground	8.03	8.05	8.00
GF pellets ⁵	4.52	4.30	4.98
CM	13.7	—	—
ESBM	—	12.1	12.5
Soybean hulls	—	1.51	1.11
Molasses ⁶	5.02	5.03	5.25
Mineral mix ⁷	1.51	1.51	1.58
CP supply from protein meals, g/kg of DMI	5.84	5.78	5.77
Nutrient composition, % of DM (or as indicated)			
CP ⁸	16.4	16.6	16.5
RDP ⁹	10.5	9.9	9.0
RUP ⁹	5.9	6.7	7.4
aNDF ⁸	31.0	29.6	29.6
ADF ⁸	22.7	21.2	21.1
EE ⁸	4.7	5.3	5.6
NEL, ⁹ Mcal/kg of DM	1.54	1.58	1.59
NEL balance, ⁹ Mcal/d	−0.6	1.3	0.4
MP balance, ^{9,10} g/d	−268	60	132
RDP balance, ^{9,10} g/d	231	11	−235
RUP balance, ^{9,10} g/d	−406	80	170
NFC ⁸	43.6	43.7	44.3
Starch ⁸	23.7	24.1	24.3
Ca ⁸	0.76	0.69	0.69
P ⁸	0.42	0.34	0.34

¹Average diet composition for adaptation and experimental periods (i.e., experiment wks 1 to 6). CM = canola meal; ESBM = extruded soybean meal; LSBM = soybean meal extruded at 149°C; HSBM = soybean meal extruded at 171°C.

²Corn silage was 39.7% DM and contained (DM basis) 6.4% CP and 33.5% NDF.

³Alfalfa haylage was 31.8% DM and contained (DM basis) 20.3% CP and 41.2% NDF.

⁴Straw-hay mixture was 86.5% DM and contained (DM basis) 10.1% CP and 65.1% NDF.

⁵Stocker Grower 14; Purina Animal Nutrition LLC. See Material and Methods for details.

⁶Liquid molasses from Westway Feed Products (Tomball, TX).

⁷Mineral/vitamin premix (Cargill Animal Nutrition, Cargill Inc., Roaring Spring, PA) contained (%; as-is basis) limestone, 36.75; dry corn distillers grains with solubles, 29.00; NaCl, 24.85; MgO (54% Mg), 4.15; Bio-Phos, 2.45; zinc sulfate, 0.96; mineral oil, 0.5; vitamin E, 0.37; manganese sulfate, 0.37; copper sulfate, 0.26; ferrous sulfate, 0.16; selenium, 0.13; vitamin A, 0.03; vitamin D3, 0.013; calcium iodate, 0.008; and cobalt carbonate, 0.005.

⁸Values calculated using the chemical analysis (Cumberland Valley Analytical Services Inc., Waynesboro, PA) of the feed ingredients and their inclusion in the diets.

⁹Estimated based on NRC (2001) and in situ data from Rock River Laboratory Inc. (Watertown, WI) for the protein meals using average DMI, MY, milk composition, and BW of the cows during the experiment.

¹⁰Metabolizable protein balance estimated based on NRC (2001) using CM and ESBM protein degradability values analyzed using in situ data from Rock River Laboratory Inc.

NDF content of CM (Table 1). Rumen-degraded protein and RUP values for CM and ESBM were determined using an in situ procedure at Rock River Laboratory Inc. (RRL; Watertown, WI; details of this procedure can be found at <https://rockriverlab.com/pages/Animal-Nutrition-Analysis.php>). Data from the NRC (2001) feed library were used for RDP and RUP contents of all other feed ingredients.

The CM diet was formulated to meet or exceed NE_L requirements and to supply MP at 5% below NRC

(2001) recommendations. In the LSBM diet, CM was replaced with LSBM on an equal CP basis. The LSBM diet was also intended to be MP-deficient at levels similar to those in the CM diet. For the HSBM diet, HSBM replaced LSBM on an equal DM basis. All feed ingredients, apart from the protein meals and soyhulls, were kept the same among diets (Table 1). It should be noted that minor differences in GreenFeed pellet intake across treatments, relative to initial intake estimates, contributed to the slight variation in dietary inclusion

rates of LSBM and HSBM in the ESBM diets. Diets were formulated considering the requirements of a Holstein cow producing 40 kg of milk/d with 3.50% milk fat and 2.89% true protein at 24.2 kg/d of DMI and a BW of 615 kg according to NRC (2001). The ESBM were produced by Fabin Bros. (Indiana, PA), whereas the CM was purchased through Cargill Animal Nutrition (Cargill Inc., Roaring Spring, PA).

Sampling and Measurements

Diet and Feed Ingredients. The amount of feed offered and refused was weighed individually and recorded daily for each cow at the time of feeding to measure daily as-fed intake during the entire experiment. The weekly DM content of the TMR and refusals were used to calculate TMR DM and consequently, averaged daily DMI for the week. Samples of concentrate feeds were collected weekly, and samples of the forages, TMR, and refusals were collected twice weekly and stored at -20°C . Feed samples were later dried for 72 h at 55°C in a forced-air oven and ground in a Wiley mill (Thomas Scientific) through a 1-mm screen for further analyses. Feed samples were composited (on an equal DM weight basis) to form one composite sample for the entire experiment and were submitted to Cumberland Valley Analytical Services (Waynesboro, PA) for wet chemistry analyses of CP (method 990.03; AOAC International, 2000), amylase-treated NDF (**aNDF**; Van Soest et al., 1991), EE (method 2003.05; AOAC International, 2006), ADF (method 973.18; AOAC International, 2000), ash (method 942.05; AOAC International, 2000), minerals (method 985.01; AOAC International, 2000), and estimated NFC. Dietary concentrations of MP, RDP, and RUP were estimated based on in situ data from RRL for CM and ESBM. Starch was analyzed as described in Hall (2009). The nutrient composition of the diets (i.e., CP, NDF, ADF, EE, ash, Ca, and P) was calculated by using the analyzed composition of the individual feed ingredients and their inclusion rate in the TMR (Table 1).

Samples of CM and ESBM were analyzed for intestinal digestibility of RUP at RRL using the 3-step procedure of Calsamiglia and Stern (1995), and digestibility coefficients used for reconstitution of the diets. Dietary supplies of NEL and MP were estimated using NRC (2001) based on average DMI, MY, milk composition, and BW of the cows during the experiment (Table 2).

Milk Production and Composition. Milk production of the cows was recorded daily at each milking (DeLaval DelPro). Milk samples were collected from 2 consecutive milkings (p.m. and a.m.) weekly during the experimental period. Aliquots of milk samples were placed in tubes with a preservative (2-bromo-2-nitropropane-1, 3-diol) and submitted to Dairy One Cooperative Inc. (Ithaca,

Table 2. Nutrient composition (% of DM or as indicated) of CM and soybean meals extruded (ESBM) at different temperatures fed during the experiment

Item	ESBM ¹		
	CM	LSBM	HSBM
CP ²	42.6	47.8	46.1
RDP, ³ % of CP	68.6	55.8	33.8
RUP, ³ % of CP	31.4	44.2	66.2
RUP intestinal digestibility, ⁴ % of RUP	81.9	98.1	98.0
aNDF ²	25.0	9.37	12.1
NDICP ⁵	0.92	0.98	0.62
ADF ²	19.0	4.42	4.70
ADICP ⁵	0.60	0.00	0.10
Ash ²	7.77	6.98	6.71
Ca ²	0.79	0.25	0.24
P ²	1.01	0.68	0.67
EE	3.78	9.11	10.18

¹LSBM = soybean meal extruded at 149°C ; HSBM = soybean meal extruded at 171°C .

²Analyzed by Rock River Laboratories Inc. (Watertown, WI) using wet chemistry methods.

³Estimated using in situ values from Rock River Laboratories Inc. assuming a passage rate of 6%/h based on NRC (2001) recommendations.

⁴Analyzed by Rock River Laboratories Inc. using the Calsamiglia and Stern (1995) method.

⁵NDICP = neutral detergent insoluble crude protein, ADICP = acid detergent insoluble crude protein.

NY) for analysis of milk fat, true protein, lactose, SCC, and MUN using infrared spectroscopy. Milk composition data were weighted for the corresponding MY during p.m. and a.m. milkings. The averaged MY during each experimental week was used to calculate milk fat, true protein, and lactose yields; ECM was calculated according to Sjaunja et al. (1991). Separate milk samples were collected weekly during the experimental period, placed in tubes without preservative, and stored at -20°C until further analysis. Samples were composited for the entire experiment and analyzed for milk FA composition of milk fat as described in Rico and Harvatine (2013).

BW, BCS, and BW Change. Cow BW was recorded twice daily upon exiting the milking parlor using a walk-through scale system (DeLaval DelPro). Body weight change was calculated as the difference between the average BW during experimental wk 6 and the average BW during wk 2 of the covariate period divided by days on study. Body condition scoring was done by the dairy manager both during the covariate period and experimental wk 6. Change in BCS was calculated as the difference between BCS during experimental wk 6 and BCS during wk 2 of the covariate period.

Ruminal Fermentation. Whole ruminal contents samples were collected from 8 ruminally cannulated cows. Three ruminally cannulated cows were assigned to each of the CM and LSBM diets, and 2 were assigned to the HSBM diet. Whole rumen contents were collected from 4

locations within the rumen during wk 5 of the experiment at 0, 2, 4, and 8 h after feeding, following the procedure described by Hristov et al. (2011). However, due to the unbalanced number of cannulated cows across treatments and to increase statistical power, whole ruminal contents samples were collected from 18 cows (6 cows per treatment) during experimental wk 5, at 4 h after feeding using the ororumenal technique (stomach tubing) as described in de Assis Lage et al. (2020). Briefly, the sampling device consisted of a 244-cm-long polyethylene orogastric tubing with a 15-mL perforated conical tube attached. An electric vacuum pump (Gast model 0823-v131q-g608nex, Septic Solutions Inc.) was used to obtain rumen contents. During the sampling event, rumen fluid was collected by placing an oral speculum in the mouth of the animal and pushing the orogastric tube down the esophagus, through the fiber mat, into the rumen. After discarding the first 200 mL to avoid saliva contamination, ~500 mL of rumen contents were collected for further analyses. Whole rumen digesta from cannulated and noncannulated cows were filtered through 2 layers of cheesecloth. The filtered rumen fluid was analyzed immediately for pH (59000–60 pH Tester, Cole-Parmer Instrument Company, Vernon Hills, IL), and aliquots were processed and analyzed for VFA and NH_3 as described in Stepanchenko et al. (2023).

Enteric Gas Emission. Two GreenFeed units (C-Lock Inc., Rapid City, SD) were permanently located in the freestall pens and available for cows to visit and take measurements of enteric gas emissions (CH_4 , CO_2 , and H_2). Briefly, a pelletized bait feed (Stocker Grower 14, Purina Animal Nutrition LLC, Shoreview, MN) was used to attract the cows to the GreenFeed units, and the weight of pellets dispensed was recorded and included in the daily DMI. Cows were allowed a maximum of 6 visits in 24 h, with 4-h intervals between visits and no more than 12 feed drops of ~33 g each per visit. Cows were identified with radiofrequency identification ear tags and were adapted to GreenFeed before the beginning of the experiment. The GreenFeed units were calibrated weekly, and CO_2 recovery tests were done monthly following the manufacturer's recommendations (<http://greenfeed.c-lockinc.com>). Emissions were measured during the covariate period and throughout the adaptation and experimental periods and averaged weekly. An average of 3.28 ± 1.40 visits/cow per day was recorded and processed during the experimental period of the study. At the end of the experiment, daily intake of pellets was added in the DMI calculation of each cow. Weekly average DMI and milk and ECM yields were used to estimate weekly averages of CH_4 and CO_2 yields (i.e., grams/kilogram of DMI) and intensity (i.e., grams/kilogram of MY or ECM).

Apparent Total-Tract Digestibility and Nitrogen Utilization. Spot urine and fecal samples (~300 mL and 500

g per sample, respectively) were collected at 0, 4, 8, and 12 h after feeding for 2 consecutive days during experimental wk 6 for estimation of N utilization and apparent total-tract digestibility of nutrients. A full description of the urine and fecal sample processing and analyses can be found in Lee et al. (2012) and Oh et al. (2013). Briefly, raw urine from each sampling was acidified with sulfuric acid, diluted 1:10, and composited by cow. Samples were then frozen at -20°C for later analysis of allantoin, uric acid, creatinine, urine urea N (UUN), and total N. Allantoin was analyzed following the procedure by Chen et al. (1992). Stanbio Laboratory (Boerne, TX) kits were used to analyze uric acid (Uric Acid Kit 1045), creatinine (Creatinine Kit 420), and UUN (Urea Nitrogen Kit 580). Total N was analyzed in freeze-dried urine samples using a Costech ECS 4010 C/N/S elemental analyzer (Costech Analytical Technologies Inc., Valencia, CA). Daily volume of excreted urine was estimated based on urinary creatinine concentration, assuming a creatinine excretion rate of 29 mg/kg of BW based on unpublished total urine collection data from Hristov et al. (2011). Daily total N, UUN, and purine derivatives excretions were calculated using the estimated urine output.

Fecal samples were oven dried at 55°C , ground through 1-mm screen in a Wiley mill (Thomas Scientific), and analyzed for DM, OM, CP, starch, NDF, and ADF. Analysis of ash was conducted by ashing the samples for 4 h at 600°C . A Mixer Mill MM 200 (Retsch GmbH, Haan, Germany) was used to pulverize a 0.5-g aliquot of fecal sample for CP analysis ($\text{N} \times 6.25$) using the Costech ECS 4010 C/N/S elemental analyzer. Samples were analyzed for aNDF, and ADF using an Ankom 200 fiber analyzer (Ankom Technology Corp., Macedon, NY) based on Van Soest et al. (1991). Starch analysis of fecal samples was according to Hall (2009). Apparent total-tract digestibility of nutrients was estimated using indigestible NDF (iNDF) as an intrinsic digestibility marker (Schneider and Platt, 1975). Fecal and TMR samples were analyzed for iNDF using a 12-d ruminal incubation in situ procedure according to Huhtanen et al. (1994), with the exception that 25- μm pore size Ankom filter bags (F57; Ankom Technology, Macedon, NY) were used for the rumen incubation (Lee et al., 2012).

In Situ Protein Degradability. Ruminal degradability of CP in CM, LSBM, and HSBM was determined in situ. Three lactating Holstein cows were used for the in situ experiment and were fed the following diet (% of DM): corn silage, 37.4; alfalfa haylage, 14.0; grass-hay mixture, 3.22; ground corn grain, 10.2; CM, 9.3; whole roasted soybeans, 7.0; cookie meal, 7.0; whole cottonseed, 5.5; molasses, 4.7; and a mineral and vitamin premix, 2.0. The experimental procedures were as described by (Cueva et al., 2023a). Samples of CM, LSBM, and HSBM were collected throughout the experiment and

composited on an equal as-is basis. Five grams of each of the protein meals (as-fed basis and without further processing) were weighed into Ankom polyester bags (10 × 20 cm nylon bags with 50 ± 10 µm porosity; Ankom Technology Corp., Macedon, NY), which were sequentially incubated in the ventral rumen for 0 (processed as the other samples but not incubated in the rumen), 2, 4, 6, 16, 24, and 48 h and simultaneously removed. Bags were incubated in triplicate for each time point and cow. Samples were presoaked in warm (39°C) water for 1 min before being placed into the rumen. Following incubation, all bags were manually washed, including the 0-h samples, with cold tap water until the water ran clean. All samples were oven dried for 72 h at 55°C, and aliquots of each bag residue were pulverized using a Mixer Mill MM 200 (Retsch GmbH, Haan, Germany) and analyzed for N using an elemental analyzer (Costech ECS 4010 C/N/S, Costech Analytical Technologies Inc., Valencia, CA). Crude protein was calculated as N × 6.25. Ruminal disappearance was calculated based on initial dry weight of the incubated sample, residue dry weight, and N concentration of the initial sample and bag residue. In situ N disappearance from the nylon bags incubated in the rumen was assumed to be degraded by rumen microorganisms. Degradation curves were fitted to a single, 3-parameter exponential rise to a maximum model (SigmaPlot 13.0; Systat Software, Chicago, IL):

$$P = a + b \times (1 - e^{-ct}),$$

where P (%) is the degraded fraction of protein meal N at time t , a (%) is the soluble fraction of N, b (%) is the potentially degradable fraction, and c (%/h) is the rate of degradation of fraction b (Ørskov and McDonald, 1979); $a + b$ was constrained to ≤100%. The in situ effective degradability (ED, %) of N was determined using the following equation (Ørskov and McDonald, 1979):

$$ED = a + b \times [c \div (c + k)],$$

where a , b , and c are as specified above and k is the rate of passage, assumed to be 0.06/h which is passage rate estimated for concentrate feeds when DMI is at 4% of BW and the diet is 50% forages (NRC, 2001).

Statistical Analysis

All data were analyzed using the MIXED procedure of SAS, version 9.4 (SAS Institute Inc., Cary, NC). Data were tested for normality using the UNIVARIATE procedure and were processed for outlier identification using the REG procedure based on an absolute studentized residual value >3. Log-transformed data were analyzed when the W statistic of the Shapiro–Wilk test was less

than 0.05 (i.e., SCC). Production and gas emission data were averaged by week and the average values were used in the statistical analysis. The individual cow average daily MY and milk compositions during each experimental week were used to calculate yields of milk fat, true protein, lactose, ECM yield, and ECM feed efficiency. These data were analyzed as repeated measures using the AR(1) covariance structure; week was the repeated term and block × treatment was the subject. Data collected during the 3-wk adaptation period were not included in the statistical analysis. The statistical models for the production data (DMI, MY, and ECM yield, milk composition, BW, feed efficiency, and ECM feed efficiency) included the fixed effects of treatment, week, parity, treatment × week, and a covariate measurement. Treatment × parity interaction was tested but removed from all final models ($P > 0.05$). Block and block × treatment were random effects. Orthogonal contrasts were used to evaluate the effect of replacing CM with ESBM (i.e., CM vs. LSBM and HSBM) and extrusion temperature of ESBM (i.e., LSBM vs. HSBM). The in situ degradability data were also analyzed using an orthogonal contrast to compare differences in the in situ N fractions between CM and ESBM and between LSBM and HSBM.

Bodyweight change, ruminal fermentation, milk FA, nutrient intake, and apparent digestibility, as well as N utilization and excretion data, were analyzed with treatment and parity as fixed effects in the statistical model. Block and block × treatment were random effects.

Statistical differences were considered significant at $P \leq 0.05$ and a trend was declared at $0.05 < P \leq 0.10$. Unless indicated otherwise, data are presented as covariate-adjusted LSM.

RESULTS AND DISCUSSION

Diet and Feed Characteristics

Substitution studies can be complex because nutrients cannot be adjusted in isolation, often requiring the addition or removal of ingredients beyond the primary ingredient of interest. Consequently, soybean hulls were added to the ESBM diets to offset the difference in inclusion rate between CM and ESBM and partially compensate for the greater NDF content of CM.

Due to higher EE concentration in ESBM relative to CM, the NEL content of the LSBM diet was 2.6% higher and that of the HSBM diet was 3.2% higher compared with the CM diet. Cows fed the CM diet were on a slightly negative energy balance, whereas those fed the ESBM diets exceeded their energy requirements by up to 3%. Both ESBM diets supplied more RUP to the cows and MP balance was positive (+2.1% and 4.6%, for the LSBM and HSBM diets, respectively), whereas

the CM diet supplied MP 9% below requirements, according to NRC (2001).

As mentioned earlier, our initial goal was to formulate the CM diet to be around 5% MP-deficient according to NRC (2001) requirements. However, an MP deficit greater than initially anticipated was observed in cows fed the CM diet (Table 1), caused partially by higher milk true protein yields (3% higher) relative to milk composition of the CM-fed cows when diets were formulated. In addition, based on the nutrient composition reported in Lage et al. (2021) and Cueva et al. (2023b), RUP concentration in CM was expected to be greater than what was observed for CM in the current experiment (as discussed in the following section). Furthermore, an MP deficit was not achieved for the ESBM diets due to a greater-than-anticipated DMI (4.8% greater, on average) and similar milk and milk components production relative to ESBM-fed cows before the beginning of the experiment. It should be also noted that RDP supply with the CM and LSBM diets met or exceeded NRC (2001) recommendations, whereas the HSBM diet supplied RDP 10% below recommendations.

Composition of Protein Meals and In Situ N Disappearance

The nutrient composition of the protein meals used in the current study is presented in Table 2. On average, the CP content of CM was 4.4 percentage units lower than that of both ESBM diets. Additionally, CM had a higher concentration of RDP and lower concentration of RUP, compared with ESBM. The HSBM had the highest RUP content among all 3 meals, which is consistent with the known effect of extrusion temperature on RUP (Giallongo et al., 2015). In this experiment, the RUP concentrations in both CM and LSBM were similar to the mean values reported by NASEM (2021) for mechanically extracted CM and ESBM (32.0% and 45.0% of CP, respectively). The intestinal digestibility of RUP, assessed using the 3-step procedure of Calsamiglia and Stern (1995), was ~16% greater for ESBM compared with CM. Canola meal had nearly 3 times greater aNDF concentration than ESBM. As expected, the EE concentration was 2.4 and 2.7 times greater in LSBM and HSBM, respectively, compared with CM.

In situ N disappearance curves for CM and both ESBM derived from the in situ subexperiment are shown in Figure 1. The in situ N degradation rate of CM was greater ($P < 0.01$) than degradation rates of ESBM. Increasing extrusion temperature (i.e., LSBM vs. HSBM) tended to decrease ($P = 0.07$) N degradation rate in ESBM. Canola meal exhibited greater ($P < 0.01$) ruminal N ED relative to the ESBM, which agrees with the greater RDP content, compared with ESBM, reported for CM in NASEM

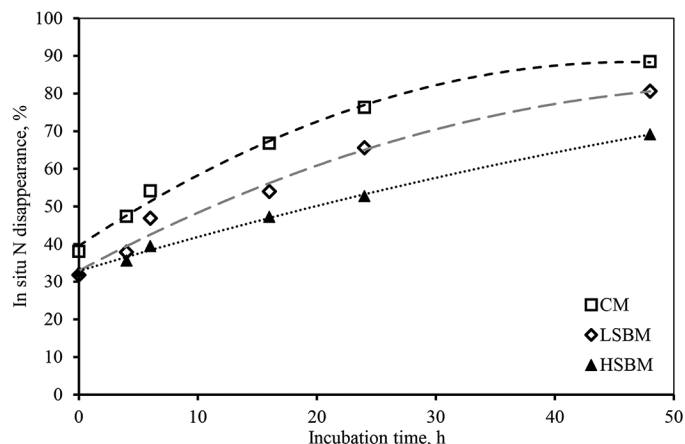


Figure 1. In situ disappearance of nitrogen from the CM and soybean meals extruded (ESBM) at different temperatures used in the experiment. CM = canola meal; LSBM = soybean meal extruded at 149°C; HSBM = soybean meal extruded at 171°C. The rate of disappearance (%/h; mean $\pm$ SE) was 4.7 ± 0.31 , 2.9 ± 0.38 , and 1.7 ± 0.31 (CM, LSBM, and HSBM, respectively). The ED (%; estimated at 6%/h rate of passage) was 63.0 ± 1.47 , 53.8 ± 1.47 , and 47.2 ± 1.49 (CM, LSBM, and HSBM, respectively). Main effect of CM versus ESBM: $P < 0.01$ for both variables. Orthogonal contrasts: CM vs. ESBM: $P < 0.01$ for both variables; LSBM vs. HSBM: $P = 0.07$ for rate of N disappearance and $P = 0.04$ for effective ruminal degradability. Regression, $P < 0.01$ for all meals; R^2 for all meals = 0.98 to 1.00. Regression lines: Dark short-dash line = CM, gray medium-dash line = LSBM, dark round-dot line = HSBM.

(2021). In addition, increasing the extrusion temperature of ESBM decreased ($P = 0.04$) ED of N in ESBM.

Based on previous research from our group and the processing characteristics of CM, specifically mechanical extrusion preceding solvent extraction, which is required due to the high oil content of canola seeds (Canola Council of Canada, 2024), the RUP content of CM was expected to be higher than the values reported in NRC (2001) and NASEM (2021). For instance, in the study by Lage et al. (2021), concentration of RUP in CM was nearly double the mean NRC (2001) value for that feed (63% vs. 32% of CP, respectively). Similarly, Cueva et al. (2023b) observed a 50% greater RUP content of CM compared with NASEM (2021) values, which aligned with RUP for CM reported earlier by Maxin et al. (2013b) and Broderick et al. (2016). As discussed by Broderick et al. (2016), the composition of CM protein fractions can vary among batches and processing plants, potentially explaining the discrepancy in CM RUP content between the current experiment and those reported in previous studies by our group, which were analyzed using the in situ method (Lage et al., 2021; Cueva et al., 2023b; 31% versus an average of 55% of CP, respectively). Discrepancies in protein composition of oilseed meals may arise not only from agronomic factors, such as cultivar and environmental conditions (Paula et al., 2017; Pierce et al., 2024), but also from methodological limitations as-

sociated with estimating degradability and digestibility of protein fractions. In particular, the in situ technique can introduce variability due to factors like physical loss of particles through the bag, inadequate incubation time, and microbial adherence to the feed residues. Such issues can affect the accuracy of RDP and RUP estimates (Hristov et al., 2019).

Results from the in situ study are consistent with analysis from RRL showing higher RDP concentration in CM compared with ESBM, as evidenced by the increased in situ ED of N from CM, relative to ESBM. Additionally, in the current study, the rates of ruminal N degradation were lower for the ESBM compared with CM. Similar findings regarding difference in ruminal N disappearance between CM and ESBM were previously reported by Lage et al. (2021) and Cueva et al. (2023b). The RUP content of LSBM (extruded at 149°C) was 4 percentage units greater than that of ESBM produced at similar extrusion temperatures as reported by Giallongo et al. (2015). However, ESBM was ~6 and 10 percentage units lower than the RUP of ESBM extruded at 161°C used in the studies by Harper et al. (2019) and Cueva et al. (2023b), respectively. Notably, the RUP concentration in HSBM (extruded at 171°C) used in this experiment was 10 percentage units higher than that of the ESBM extruded at the same temperature reported in Giallongo et al. (2015). Data from both the Giallongo et al. (2015) report and the current study demonstrate clear reductions in the in situ N ED and degradation rate with increasing extrusion temperature of ESBM. Furthermore, the results of the current experiment align with findings from Isenberger et al. (2012), showing a linear increase ($R^2 = 0.99$) in the RUP content of ESBM with rising temperatures, as illustrated in Figure 1. Differences in the in situ ED of N between LSBM and HSBM were 6 percentage units in the current experiment versus a 12 percentage unit difference in the study of Giallongo et al. (2015) for ESBM produced at the same extrusion temperatures. This discrepancy can be attributed to variability of in situ degradability values because the meals were produced in the same facility. Factors such as the absence of chewing, assumptions regarding retention time, potential physical loss of feed particles through bag pores, and inaccurate quantification of microbial contamination on undigested residues can significantly affect in situ estimates, as discussed by Huhtanen and Ahvenjärvi (2022).

Production Effects

Treatments did not affect DMI, MY, ECM yield, and feed efficiency metrics in this study (Table 3). Milk fat concentration decreased ($P = 0.04$) and milk fat yield tended to decrease ($P = 0.08$) when CM was replaced with ESBM in the diet. There was no effect of extrusion

temperature of ESBM on milk fat. Milk true protein content tended to decrease ($P = 0.10$) when CM was replaced with ESBM and by increasing the extrusion temperature. However, milk true protein (MTP) yield was not unaffected by treatments. Milk urea N was increased ($P < 0.01$) by ESBM compared with CM, whereas increasing the extrusion temperature of ESBM had no effect on MUN. Treatments had no effect on BW and BCS.

In a meta-analysis, Huhtanen et al. (2011) reported increased DMI and MY in lactating dairy cows fed CM relative to SBM. Furthermore, more recent research has reported that MY increased (Paula et al., 2020; Lage et al., 2021) when CM replaced SSBM in lactating cow diets. Cueva et al. (2023b) described similar DMI, MY, and ECM yield in dairy cows fed CM and ESBM at equal CP inclusion rates. Lage et al. (2021) reported increased DMI but similar MY and ECM for cows fed CM compared with ESBM in a crossover experiment. In that study, both the CM and ESBM diets were deficient in Met, with the ESBM diet exhibiting the largest deficiency. Huhtanen et al. (2011) suggested that replacing SBM with CM in lactating cow diets could enhance DMI by improving the supply of EAA, particularly Met, which supports greater MY and prompts cows to increase DMI to meet higher energy demands. In the current study, cows fed the CM diet did not have lower milk production than the ESBM cows, despite being in a negative MP balance, approximately -9% of NRC (2001) requirements. In addition, and after considering the lack of response to greater MP supply by the ESBM diets, it is plausible that NRC (2001) underpredicts lactational performance of dairy cows fed MP-deficient diets (Lee et al., 2011). It should also be noted that MP supply estimations are highly dependent on the input values for protein fractions of feedstuffs. Therefore, variability in feed evaluation, such as the limitations of in situ methods discussed herein, could result in inaccurate estimates of MP supply and confound interpretation of production responses.

Giallongo et al. (2015) did not report production benefits when substituting LSBM with HSBM in the diet of lactating cows. This aligns with previous studies that reported inconsistent MY and DMI responses when heat-treated SBM was included in dairy cow diets (Broderick, 1986; Broderick et al., 1990). The estimated higher RUP supply by the HSBM diet compared with LSBM, however, did not result in performance benefits, likely due to an oversupply of RUP and MP relative to cow requirements or an undersupply of RDP for utilization by ruminal microbes (discussed later in this section). This hypothesis, however, is not supported by the trend for decreased milk protein concentration by the ESBM diets and specifically the HSBM diet.

A meta-analysis conducted by Martineau et al. (2013) reported no significant change in milk fat content when

Table 3. Dry matter intake and milk production variables in dairy cows fed diets containing CM soybean meals extruded (ESBM) at different temperatures fed during the experiment

Item	Treatment ¹			SEM ³	P-value ²	
	CM	LSBM	HSBM		CM vs. ESBM	LSBM vs. HSBM
DMI, kg/d	27.3	27.2	26.7	0.65	0.81	0.74
MY, kg/d	43.4	44.0	44.8	1.18	0.39	0.52
Feed efficiency, ⁴ kg/kg	1.59	1.61	1.66	0.040	0.28	0.30
Milk fat, %	4.07	3.73	3.79	0.119	0.04	0.71
Milk fat yield, kg/d	1.75	1.60	1.66	0.062	0.08	0.41
ECM, ⁵ kg/d	42.7	41.3	40.8	1.24	0.22	0.76
ECM feed efficiency, ⁶ kg/kg	1.55	1.50	1.51	0.041	0.38	0.86
Milk true protein, %	3.00	2.96	2.85	0.045	0.10	0.10
Milk true protein yield, kg/d	1.28	1.26	1.25	0.029	0.37	0.80
Milk lactose, %	4.80	4.86	4.86	0.061	0.30	0.99
Milk lactose yield, kg/d	2.09	2.15	2.11	0.065	0.44	0.47
MUN, mg/dL	10.0	11.5	11.3	0.30	0.001	0.65
SCC × 10 ³ cells/mL ⁷	67.7 (1.67)	27.5 (1.42)	109.3 (1.65)	25.7 (0.07)	0.08	0.02
BCS	2.95	3.13	3.05	0.098	0.11	0.47
BCS change ⁸	0.09	0.03	0.11	0.072	0.83	0.37
BW, kg	672	651	654	10.4	0.11	0.81
BW change, ⁹ kg/d	0.28	0.01	0.01	0.21	0.24	0.96

¹Treatments were: CM = canola meal diet, LSBM = diet with soybean meal extruded at 149°C, and HSBM = diet with soybean meal extruded at 171°C. Means are covariate-adjusted LSM.

²Orthogonal contrasts: CM vs. ESBM = CM vs. diets formulated using ESBM and LSBM vs. HSBM = soybean meal extruded at 149°C versus 171°C.

³Largest SEM published in table; n = 255 for all variables except DMI and MY and feed efficiency, for which n = 262; (n represents number of observations used in the statistical analysis).

⁴Milk yield ÷ DMI.

⁵Energy-corrected milk (kg/d) = kg of milk × [(38.3 × % fat × 10 + 24.2 × % true protein × 10 + 16.54 × % lactose × 10 + 20.7) ÷ 3,140] (Sjaunja et al., 1991).

⁶ECM yield ÷ DMI.

⁷Statistical analysis was performed on log-transformed data; actual data are shown in parentheses following the log-transformed values.

⁸BCS change was: (average BCS during experimental wk 6 – average BCS covariate wk 2).

⁹BW change: (average BW during experimental wk 6 – average BW covariate wk 2) ÷ days on study.

CM substituted SSBM in lactating dairy cow diets. This finding is similar to the results reported by Lage et al. (2021), where an equal CP replacement of CM with ESBM did not affect milk fat concentration. In contrast, Cueva et al. (2023b) observed increased milk fat content and yield in cows fed ESBM compared with CM. It is noted, however, that the addition of free canola oil to the CM diet in the Cueva et al. (2023b) experiment may have affected rumen biohydrogenation of FA and milk fat synthesis (Bauman and Griinari, 2003). In the present study, the concentration of aNDF in the ESBM diets was lower than in the CM diet, which may have also partially affected rumen fermentation and consequently milk fat concentration (Ipharraguerre and Clark, 2003).

The meta-analysis by Huhtanen et al. (2011) did not indicate an increase in MTP concentration in diets containing CM compared with SSBM diets. However, Martineau et al. (2013) reported an increase in MTP yield due to higher MY in cows fed CM diets. Previous studies from our research group (Lage et al., 2021; Cueva et al., 2023b), did not describe significant effects of replacing CM with ESBM on MTP concentration. In the current study, cows fed the ESBM diets tended to

have lower MTP concentrations compared with the CM diet, and MTP concentration tended to be 4% lower for HSBM than LSBM. As discussed by Reynal and Broderick (2005), an increase in dietary RUP at the expense of RDP may limit available protein supply to the rumen, hindering microbial growth and ruminal fermentation in general. Thus, the lower MTP concentration in cows fed the ESBM diets may have resulted from a decreased supply of RDP, which was particularly true for the HSBM diet (Table 1). The milk protein effect for the HSBM diet is also in line with the decreased total-tract CP digestibility for that diet.

The greater MUN concentrations observed in cows fed the ESBM diets compared with those fed CM were likely due to the higher RUP and overall MP supply provided by the ESBM diets. Consistent with these findings, both Cueva et al. (2023b) and Lage et al. (2021) reported increased MUN concentrations in cows fed ESBM relative to CM. In Lage et al. (2021), elevated MUN was observed despite a greater supply of Met in the ESBM diets, which did not translate into increased MTP yield. This milk protein response is similar to the present study, where model estimates predicted comparable

Met supplies between CM and ESBM diets (49 vs. 52 g/d, respectively). As discussed by Lage et al. (2021), the efficiency of converting metabolizable AA into milk protein is not constant across individual AA and depends on both MP supply and demand. Therefore, differences in MP supply between diets may have influenced how Met was utilized by the mammary gland. Furthermore, as inferred from data reported by Nousiainen et al. (2004), elevated RUP supply from ESBM without a corresponding increase in milk protein synthesis is likely to lead to increased AA catabolism and, consequently, greater MUN concentrations.

Milk FA

Milk FA data are shown in Table 4. Concentrations of most of the FA were not affected by treatment, except that 13:0 iso, 14:0 iso, and 16:0 iso were decreased or tended to be decreased ($P \leq 0.10$) by ESBM compared with the CM diet. There was no effect of treatment on individual or total *trans* FA. Overall, diets fed in the current experiment did not affect the sum of de novo, preformed, and odd- and branched-chain FA (OBCFA). In contrast, the sum of mixed milk FA (16:0 and 16:1 *cis*-9) tended to be greater ($P = 0.10$) in milk fat of cows fed ESBM when compared with CM. No differences were observed between the LSBM and HSBM diets in FA profile of milk fat in the current experiment.

The protein meals used in the experiment were not analyzed for FA composition. In the study by Lage et al. (2021) cows fed CM and ESBM had similar milk FA profiles, with the exception that concentration of PUFA was decreased in cows fed an ESBM diet, relative to CM, an effect that the authors attributed the inclusion of supplemental canola oil in the CM diet relative to ESBM. The tendency for an increased sum of mixed milk FA in milk fat of cows fed the ESBM relative to the CM diet was mainly due to the numerical trend for greater concentration of 16:0 for the former diets. The numerically greater concentration of 16:0 FA in milk from cows fed ESBM compared with CM is likely due to the higher 16:0 content in ESBM, resulting in greater intake and subsequent transfer of 16:0 into milk fat. In the conditions of the current experiment, changes in milk FA composition due to replacing CM with ESBM or increasing extrusion temperature were minimal, suggesting that under these dietary conditions, substitution of protein meals had a limited effect on milk fat FA profile. However, it should be noted that ESBM had a greater EE concentration and a higher proportion of *cis*-9, *cis*-12 18:2 FA in its profile, whereas CM had lower EE but a greater proportion of C18:1 FA. As noted by Daley et al. (2020), the presence of free FA in the rumen, particularly *cis*-9, *cis*-12 18:2, has been consistently as-

sociated with negative effects on milk fat synthesis in dairy cows. In this context, the greater concentration of free FA, including linoleic acid, from ESBM may have contributed to the overall lower milk fat concentration in cows fed ESBM when compared with cows fed CM.

Enteric Gas Emission

Treatments did not affect enteric emission of CH₄, H₂, and CO₂ in the current study (Table 5). Methane emission yield (expressed per unit of DMI) and intensity (expressed in relation to MY and ECM yield) were also similar among treatments.

We initially hypothesized that the increased EE concentrations and slightly lower NDF content of ESBM diets would lead to decreased CH₄ production, compared with the CM diet. The CH₄-inhibiting effects of increasing dietary EE concentration have been extensively studied and confirmed (Martins et al., 2024). The lack of effect of ESBM diets on CH₄ production in the current study is consistent with the report by Harper et al. (2019), where the replacement of SSBM (1.3% oil) with ESBM (8.6% oil) did not affect any of the CH₄ emission metrics. In the study by Lage et al. (2021), increased CH₄ yield was reported in cows fed ESBM relative to CM. However, this difference was attributed to variations in ruminal OM digestibility between CM and ESBM diets, and their associated effects on DMI, rather than being a direct fermentation-mediated response attributable to the protein meals. In the meta-analysis conducted by Grainger and Beauchemin (2011), a linear relationship between dietary fat and enteric CH₄ emission was observed, indicating that an increase of 1% of DM in dietary fat resulted in a decrease of 1g CH₄ yield. Applying the findings of this meta-analysis to the average dietary EE contents used in the current study, it was expected that the ESBM diets would lead to a reduction in CH₄ yield of slightly below 1 g of DMI compared with CM diet, which would be of no biological or statistical relevance.

Rumen Fermentation

Ruminal pH was not affected by treatments (Table 6). Total VFA concentrations were not different between CM and the ESBM diets or between HSBM compared with LSBM. There was no effect of replacing CM with ESBM on molar proportions of VFA. Increasing the processing temperature of ESBM, however, resulted in decreased ($P = 0.04$) propionate, increased ($P = 0.01$) acetate, and a trend for increased ($P = 0.10$) isobutyrate proportions when compared with the LSBM diet. The acetate-to-propionate ratio was not affected by replacement of CM with ESBM but increased ($P = 0.03$) in HSBM compared

Table 4. Fatty acid (FA) composition (g/100 g of total FA) of milk fat in dairy cows fed diets containing CM or soybean meals extruded (ESBM) at different temperatures fed during the experiment

Item	Treatment ¹			SEM ³	P-value ²	
	CM	LSBM	HSBM		CM vs. ESBM	LSBM vs. HSBM
4:0	4.59	4.37	4.35	0.134	0.15	0.91
6:0	2.28	2.18	2.18	0.066	0.17	0.98
8:0	1.20	1.14	1.15	0.041	0.30	0.92
10:0	2.63	2.56	2.57	0.113	0.65	0.96
<i>cis</i> -9 10:0	0.23	0.22	0.22	0.011	0.48	0.79
11:0	0.055	0.055	0.053	0.006	0.87	0.77
12:0	2.95	2.91	2.90	0.134	0.79	0.99
iso-13:0	0.021	0.015	0.017	0.002	0.02	0.35
anteiso-13:0	0.058	0.059	0.057	0.004	0.96	0.71
13:0	0.092	0.094	0.091	0.007	0.91	0.76
iso-14:0	0.057	0.051	0.049	0.004	0.10	0.70
14:0	10.0	10.4	10.4	0.25	0.27	0.86
anteiso-15:0	0.35	0.34	0.33	0.008	0.22	0.64
<i>cis</i> -9 14:0	0.81	0.90	0.84	0.063	0.40	0.51
15:0	0.90	0.92	0.91	0.048	0.80	0.85
iso-16:0	0.15	0.14	0.13	0.008	0.10	0.33
16:0	27.4	28.5	28.2	0.60	0.13	0.72
iso-17:0	0.28	0.27	0.27	0.007	0.17	0.80
<i>cis</i> -9 16:1	1.17	1.31	1.23	0.065	0.17	0.35
anteiso-17:0	0.37	0.36	0.35	0.010	0.20	0.69
17:0	0.48	0.46	0.47	0.013	0.36	0.73
<i>cis</i> -9 17:1	0.29	0.28	0.27	0.015	0.22	0.76
18:0	11.4	11.0	11.1	0.38	0.41	0.64
<i>trans</i> -4 18:1	0.023	0.019	0.021	0.002	0.15	0.63
<i>trans</i> -5 18:1	0.016	0.013	0.015	0.002	0.53	0.45
<i>trans</i> -6,8 18:1	0.42	0.39	0.41	0.026	0.49	0.51
<i>trans</i> -9 18:1	0.34	0.32	0.33	0.017	0.53	0.64
<i>trans</i> -10 18:1	1.20	0.95	1.23	0.260	0.72	0.45
<i>trans</i> -11 18:1	1.34	1.22	1.25	0.086	0.30	0.79
<i>trans</i> -12 18:1	0.71	0.68	0.68	0.029	0.36	0.79
<i>trans</i> -15 18:1	0.44	0.42	0.42	0.016	0.24	0.86
<i>cis</i> -9 18:1	19.3	19.4	19.2	0.497	0.92	0.71
<i>cis</i> -11 18:1	0.67	0.67	0.74	0.048	0.48	0.28
<i>cis</i> -12 18:1	0.67	0.61	0.62	0.038	0.29	0.82
<i>cis</i> -9, <i>cis</i> -12 18:2	2.82	2.79	2.84	0.139	0.97	0.80
<i>cis</i> -6, <i>cis</i> -9, <i>cis</i> -12 18:3	0.023	0.023	0.024	0.003	0.95	0.65
20:0	0.131	0.128	0.129	0.004	0.57	0.84
<i>cis</i> -9, <i>cis</i> -12, <i>cis</i> -15 18:3	0.452	0.451	0.465	0.016	0.75	0.50
<i>cis</i> -11 20:1	0.078	0.077	0.080	0.003	0.92	0.39
<i>cis</i> -9, <i>trans</i> -10 CLA	0.016	0.013	0.013	0.002	0.13	0.79
<i>cis</i> -9, <i>trans</i> -11 CLA	0.608	0.589	0.584	0.041	0.64	0.91
n-6 20:2	0.028	0.027	0.025	0.003	0.37	0.48
n-6 20:3	0.121	0.114	0.118	0.007	0.46	0.71
n-3 20:3	0.006	0.006	0.010	0.001	0.24	0.01
n-6 20:4	0.138	0.144	0.138	0.005	0.59	0.34
n-3 20:5	0.028	0.028	0.028	0.002	0.72	0.52
22:00	0.040	0.036	0.037	0.003	0.31	0.92
<i>cis</i> -13 22:1	0.005	0.003	0.007	0.001	0.78	0.02
24:00	0.019	0.018	0.016	0.002	0.26	0.58
Others						
Total <i>trans</i> FA	5.14	4.61	4.97	0.399	0.43	0.49
ΣDe novo ⁴	24.8	24.7	24.6	0.56	0.88	0.86
ΣPreformed ⁵	41.8	40.8	41.2	0.95	0.40	0.71
ΣMixed	28.6	29.8	29.5	0.62	0.10	0.64
ΣOBCFA ⁶	2.98	2.91	2.88	0.080	0.42	0.80

¹Treatments were: CM = canola meal diet, LSBM = diet with soybean meal extruded at 149°C, and HSBM = diet with soybean meal extruded at 171°C.

²Orthogonal contrasts: CM vs. ESBM = CM vs. diets formulated using ESBM and LSBM vs. HSBM = soybean meal extruded at 149°C vs. 171°C.

³Largest SEM published in table; n = 45 (n represents number of observations used in the statistical analysis).

⁴De novo FA (<C16) are synthesized by the mammary gland.

⁵Preformed FA (>C16) originate primarily from extraction from plasma.

⁶Odd- and branched-chain FA. Sum of C11:0, iso C13:0, anteiso C13:0, C13:0, iso C14:0, iso C15:0, anteiso C15:0, C15:0, iso C16:0, iso C17:0, anteiso C17:0, C17:0, and C17:1 *cis*-9.

Table 5. Enteric gas emission in dairy cows fed diets containing CM or soybean meals extruded (ESBM) at different temperatures fed during the experiment

Item	Treatment ¹			SEM ³	P-value ²	
	CM	LSBM	HSBM		CM vs. ESBM	LSBM vs. HSBM
CH ₄						
CH ₄ , g/d	399	380	387	13.1	0.49	0.77
CH ₄ per DMI, g/kg	14.6	14.7	15.0	0.44	0.69	0.55
CH ₄ per MY, g/kg	9.29	8.98	9.00	0.323	0.49	0.95
CH ₄ per ECM ⁴ yield, g/kg	9.39	9.54	9.50	0.301	0.67	0.80
CO ₂ , g/d	13,911	13,651	13,505	297	0.36	0.75
H ₂ , g/d	1.45	1.54	1.47	0.13	0.75	0.92

¹Treatments were: CM = canola meal diet, LSBM = diet with soybean meal extruded at 149°C, and HSBM = diet with soybean meal extruded at 171°C. Means are covariate-adjusted LSM.

²Orthogonal contrasts: CM vs. ESBM = CM vs. diets formulated using ESBM and LSBM vs. HSBM = soybean meal extruded at 149°C vs. 171°C.

³Largest SEM published in table; n = 206; (n represents number of observations used in the statistical analysis).

⁴Energy-corrected milk (kg/d) = kg of milk × [(38.3 × % fat × 10 + 24.2 × % true protein × 10 + 16.54 × % lactose × 10 + 20.7) ÷ 3,140]; Sjaunja et al. (1991).

with LSBM. There was no effect on ruminal concentration of NH₃ among treatments.

Previous experiments did not report differences in ruminal individual or total VFA concentrations when CM replaced SSBM (Brito and Broderick, 2007; Paula et al., 2018) or ESBM (Giallongo et al., 2015) in the diet of lactating dairy cows. In our previous study, replacing CM with ESBM led to increased total VFA concentrations (Cueva et al., 2023b). In that experiment, the difference was attributed to the potentially negative effect of free canola oil supplementation on ruminal microbial activity in the CM relative to the ESBM diet. In the present study, total VFA concentration was numerically lower in cows fed the HSBM diet compared with those fed the LSBM diet, contrasting with findings by Giallongo et al. (2015), where no differences in ruminal fermentation parameters were observed when cows were fed a LSBM or a HSBM produced at the same extrusion temperatures used for the current study. As mentioned earlier, RDP may have limited microbial growth for the HSBM diet in the current experiment, leading to reduced OM degradation (discussed in the “Apparent Total-Tract Digestibility” section) and VFA production (Reynal and Broderick, 2005). These observations, along with an assumed impaired ruminal starch degradation due to limited RDP, could help explain the reduced molar proportions of ruminal propionate and greater acetate-to-propionate ratio in cows fed the HSBM diet when compared with LSBM in the current study. Furthermore, the observed tendency for an increased concentration of isobutyrate in cows fed the HSBM diet, may suggest a reduced utilization of this VFA for the synthesis of branched-chain AA, likely due to decreased microbial protein synthesis (Broderick et al., 2002). In addition, previous research suggests that free AA and peptides may serve as necessary building blocks for optimal microbial growth (Firkins et al., 2007).

Without an adequate RDP supply in the rumen, there is potentially decreased starch digestibility, as may have been the case in cows fed HSBM in the current experiment. It should be noted that total VFA concentrations in the current experiment were relatively low, which may have been related to the use of the ororumenal sampling technique to collect rumen fluid samples. The similarities in ruminal NH₃ concentration across treatments raise a challenging question. Typically, lower RDP supply would lead to reduced ruminal NH₃ concentrations due to limited N availability. This raises the possibility that the in situ estimates used in the current study may have introduced error in the estimates of RDP and RUP for the protein meals, potentially overestimating the magnitude of RDP deficiency in cows fed the HSBM diet. In this case, the actual difference in RDP supply among treatments may have been smaller than suggested by the NRC (2001) model, which could help explain the similar ruminal NH₃ concentrations observed. This possibility should be considered when interpreting data from the current manuscript.

Apparent Total-Tract Digestibility

Nutrient intake during the fecal and urine sample collection week (i.e., experimental wk 6) and apparent total-tract digestibility data are presented in Table 7. Treatments did not affect intake of dietary nutrients in the current study. Replacement of CM with ESBM did not affect digestibility of DM and OM. However, replacement of LSBM with HSBM decreased ($P \leq 0.04$) apparent total-tract digestibility of DM and OM. There was no effect of treatments on digestibility of NDF and ADF in this study. Dietary replacement of CM with ESBM did not affect apparent total-tract digestibility of CP and starch, whereas cows fed HSBM had decreased

Table 6. Ruminal fermentation variables in dairy cows fed diets containing CM or soybean meals extruded (ESBM) at different temperatures fed during the experiment¹

Item	Treatment ²				P-value ³	
	CM	LSBM	HSBM	SEM ⁴	CM vs. ESBM	LSBM vs. HSBM
pH	6.49	6.79	6.40	0.156	0.55	0.15
Total VFA, mM	101.3	105.6	92.2	8.73	0.74	0.12
VFA (mol %)						
Acetate	59.6	57.0	61.0	1.10	0.61	0.01
Propionate	21.6	24.9	21.2	1.15	0.31	0.05
Butyrate	13.6	13.1	13.3	0.76	0.68	0.91
Isobutyrate	0.87	0.95	0.83	0.05	0.76	0.10
Valerate	2.39	2.58	1.88	0.38	0.64	0.09
Isovalerate	1.65	1.43	1.72	0.11	0.71	0.15
Acetate:propionate	2.82	2.34	2.98	0.189	0.48	0.03
NH ₃ , mM	5.90	6.91	8.71	1.00	0.14	0.23

¹Samples analyzed were collected from 18 cows using the ororuminal sampling technique once a day and from 9 ruminally cannulated cows 4 times per day during experimental wk 5.

²Treatments were: CM = canola meal diet, LSBM = diet with soybean meal extruded at 149°C, and HSBM = diet with soybean meal extruded at 171°C.

³Orthogonal contrasts: CM vs. ESBM = CM vs. diets formulated using ESBM and LSBM vs. HSBM = soybean meal extruded at 149°C vs. 171°C.

⁴Largest SEM published in table; n = 43; (n represents number of observations used in the statistical analysis).

($P \leq 0.03$) apparent digestibilities of these nutrients, compared with LSBM.

In our previous study, replacing CM with ESBM led to increased ADF digestibility, although apparent total-tract digestibilities of other nutrients were not affected (Cueva et al., 2023b). In that study, the increase in ADF digestibility was attributed to the higher availability of rapidly digestible nonforage fiber from soyhulls in the ESBM diet, along with a potential negative effect on fiber digestibility arising from the inclusion of canola oil in the CM diet. In the Lage et al. (2021) report, apart from increased CP digestibility in the ESBM diet, no differences in digestibility were observed between CM and ESBM. In that experiment, the authors attributed the greater CP digestibility to the increased intestinal digestibility of ESBM compared with CM. A similar numerical trend was observed between CM and LSBM in the current study; however, HSBM had the lowest total-tract CP digestibility among all treatments despite being similar to LSBM in intestinal RUP digestibility. This is inconsistent with the *in vitro* estimates of RUP digestibility, which were generated using the 3-step procedure of Calsamiglia and Stern (1995). Although this method is widely used and sensitive to detecting heat damage of proteins, it should be noted that the extrusion temperature for HSBM in the current study (171°C) exceeded those tested in the original method validation. In contrast, the temperature applied to LSBM (149°C) was within the tested range. It is therefore possible that intestinal digestibility of RUP, particularly for HSBM, may not have been accurately analyzed by the Calsamiglia and Stern (1995) procedure.

Furthermore, digestibility was assessed using composite samples and a single *in vitro* run per treatment, which may have limited our ability to detect treatment-level variability. These factors should be considered when interpreting the discrepancy between *in vitro* RUP digestibility and the *in vivo* total-tract CP digestibility results.

Giallongo et al. (2015) reported increases in DM, OM, and ADF digestibilities when feeding HSBM relative to LSBM. Although ruminal fibrolytic bacteria increased in HSBM-fed cows in the study by Giallongo et al. (2015), the authors could not completely explain the exact mechanism behind the increase in total-tract DM and OM digestibility reported for HSBM versus LSBM. In the present study, the reductions in DM, OM, starch, and CP digestibility observed in cows fed HSBM, compared with LSBM, could be largely attributed to the former meal's lower RDP concentrations. Reynal and Broderick (2005) previously demonstrated decreases in both apparent and true OM digested in the rumen as well as total-tract OM digestibility in cows fed diets with RDP contents below 10.6% of DM. In that study, the authors attributed the low OM digestibility effects to decreased CP digestibility and diminished NDF digestibility, originating from insufficient RDP for supporting ruminal microbial activity. Similarly, Lee et al. (2011) reported decreased total-tract digestibilities of most nutrients in cows fed RDP- and MP-deficient diets. As mentioned previously, given the comparable *in vitro* intestinal RUP digestibilities between HSBM and LSBM, it is possible that the 3-step procedure did not fully capture the effects of the higher heat applied to HSBM (171°C). Therefore,

Table 7. Intake and apparent total-tract digestibility of nutrients in dairy cows fed diets containing CM or soybean meals extruded (ESBM) at different temperatures fed during the experiment

Item	Treatment ¹			SEM ³	P-value ²	
	CM	LSBM	HSBM		CM vs. ESBM	LSBM vs. HSBM
Intake, ⁴ kg/d						
DM	25.3	26.8	26.6	1.10	0.29	0.90
OM	23.7	25.1	25.0	1.04	0.29	0.92
NDF	8.16	8.19	8.61	0.350	0.57	0.40
ADF	4.96	4.95	4.98	0.210	0.98	0.93
CP	4.14	4.44	4.39	0.180	0.22	0.83
Starch	5.98	6.45	6.46	0.264	0.15	0.98
Apparent total-tract digestibility, %						
DM	69.8	70.5	68.5	0.52	0.64	0.01
OM	71.0	72.1	70.5	0.52	0.85	0.04
NDF	46.1	45.0	45.0	0.85	0.32	0.94
ADF	41.2	40.9	40.3	0.97	0.57	0.67
CP	76.6	77.5	75.5	0.67	0.85	0.03
Starch	98.2	98.3	98.0	0.09	0.69	0.03

¹Treatments were: CM = canola meal diet, LSBM = diet with soybean meal extruded at 149°C, and HSBM = diet with soybean meal extruded at 171°C.

²Orthogonal contrasts: CM vs. ESBM = CM vs. diets formulated using ESBM and LSBM vs. HSBM = soybean meal extruded at 149°C vs. 171°C.

³Largest SEM published in table; n = 43; (n represents number of observations used in the statistical analysis).

⁴DM intake reported is during the fecal collection period (experimental wk 6) for the digestibility analysis.

the lower total-tract CP digestibility observed for HSBM may have resulted from reduced protein availability due to excessive heat, thereby affecting both ruminal and intestinal digestion. This likely also contributed to the observed reduction in total-tract OM digestibility. These processes, however, did not influence the total-tract digestibility of fiber fractions in the current study.

Nitrogen Utilization

Excretion (as grams per day or as percent of N intake) of urinary, fecal, and total excreta N and milk N secretion were not affected by treatments (Table 8). In contrast, excretion (both as grams per day and as percent of intake) of UUN was increased ($P = 0.04$) by the ESBM diets, compared with CM. Fecal N excretion (as percent of intake) was increased ($P = 0.05$) by replacement of LSBM with HSBM. Treatment did not affect urinary excretion of individual or total purine derivatives.

In the current experiment, it is possible that decreased ruminal RDP may have reduced ruminal starch digestibility, thereby limiting NH_3 utilization, increasing the ruminal NH_3 pool, and subsequently elevating MUN and UUN levels for cows fed the HSBM diet compared with LSBM. However, the difference in total-tract starch digestibility between HSBM and LSBM was only 0.3 percentage units, suggesting that changes in UUN due to starch digestibility are minimal. Additionally, the observed increases in UUN excretion and similar MTP yields observed in cows fed ESBM compared with CM in the present study corresponds with other studies from our

research group (Lage et al., 2021; Cueva et al., 2023b) which have also observed that replacing CM with ESBM elevates UUN concentrations. Additionally, the increase in fecal N excretion observed in cows fed HSBM compared with those on LSBM is related to the reduced CP digestibility in the HSBM diet, which as explained in the “Apparent Total-Tract Digestibility” section, may be related to compromised CP fermentation in the rumen and a potential effect of heat treatment on HSBM RUP digestibility that may have not been captured by the in vitro procedure. It should be noted that unaccounted N averaged 19% of intake (~132 g/cow per day), with average N retention previously estimated at 38.5 g/d (Spanghero and Kowalski, 2021) in dairy cows, thus indicating values above expected typical retention in N balance studies. As noted by Hristov et al. (2019), substantial N losses can occur during collection, processing, and analysis of urine and feces, which may compromise the accuracy of total-tract N digestion and body N balance measurements and should be considered when interpreting results from the current study.

CONCLUSIONS

Replacing CM with ESBM on an equal CP basis, did not influence DMI, MY, ECM, feed, or ECM feed efficiency. In addition, substituting LSBM with HSBM in the diet did not affect DMI, MY, ECM, feed or ECM feed efficiency. Cows fed the CM diet had a higher milk fat concentration and tended to have a higher milk fat yield, possibly due to the greater nonforage NDF content

Table 8. Nitrogen utilization and purine derivative (PD) excretion in dairy cows fed diets containing CM or soybean meals extruded (ESBM) at different temperatures fed during the experiment

Item	Treatment ¹			SEM ³	P-value ²	
	CM	LSBM	HSBM		CM vs. ESBM	LSBM vs. HSBM
N intake, g/d	674	706	706	27.2	0.33	0.99
N excretion or secretion, g/d						
Urine N	194	204	207	20.9	0.68	0.92
UUN ⁴	107	147	158	17.1	0.02	0.63
Fecal N	160	159	170	7.37	0.64	0.27
Total excreta N	355	361	378	21.1	0.59	0.59
Milk N	191	200	189	8.63	0.75	0.27
As % of N intake						
Urine N	28.4	28.8	28.8	3.22	0.92	0.99
UUN	15.9	21.1	22.1	2.25	0.04	0.75
Fecal N	23.7	22.6	24.3	0.66	0.65	0.05
Total excreta N	52.1	51.5	53.4	3.19	0.93	0.66
Milk N	28.8	28.5	26.9	1.37	0.51	0.40
Unaccounted N	18.7	21.3	17.0	3.53	0.93	0.43
Urine output, kg/d	20.0	22.1	21.0	2.31	0.55	0.74
Urinary PD excretion, mmol/d						
Allantoin	540	516	514	65.2	0.71	0.98
Uric acid	75.5	78.7	82.7	10.4	0.66	0.78
Total PD	616	594.8	599	71.6	0.80	0.96

¹Treatments were: CM = canola meal diet, LSBM = diet with soybean meal extruded at 149°C, and HSBM = diet with soybean meal extruded at 171°C.

²Orthogonal contrasts: CM vs. ESBM = CM vs. diets formulated using ESBM and LSBM vs. HSBM = soybean meal extruded at 149°C vs. 171°C.

³Largest SEM published in table; n = 43; (n represents number of observations used in the statistical analysis).

⁴UUN = urinary urea nitrogen.

and higher RDP in that meal, which may have promoted ruminal fermentation compared with the ESBM diets, or due to higher overall total-tract nutrient digestibility when compared with the HSBM diet. Replacement of CM with ESBM and increasing extrusion temperature had no effect on daily enteric CH₄ emission and emission intensity. Overall, substituting CM with ESBM on an equal CP basis in the diet does not affect lactational performance of dairy cows, except for a decrease in milk fat concentration. Increasing the extrusion temperature from 149°C to 171°C during ESBM production had no apparent advantage in terms of production benefits and may have resulted in RDP deficiency that may have inhibited microbial protein synthesis in the rumen of dairy cows.

NOTES

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Nonstandard abbreviations used: ADICP = acid detergent insoluble crude protein; aNDF = amylase-treated NDF; CM = canola meal; ED = effective degradability; EE = ether extract; ESBM = extruded soybean meal; FA = fatty acid; HSBM = high-temperature ESBM; iNDF = indigestible NDF; LSBM = low-temperature ESBM; MTP = milk true protein; MY = milk yield; NDICP = neutral detergent insoluble crude protein; OBCFA = odd- and branched-chain FA; PD = purine derivative; RRL = Rock River Laboratory; SBM = soybean meal; SSBM = solvent-extracted soybean meal; UUN = urinary urea N.

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